A Dataset Construction Details

A.1 Privacy Category Taxonomy

The taxonomy review group consisted of two master's students in computer science, two doctoral students with master's degrees in computer science, and one doctoral student in social welfare.

High-level category	**Subcategory**	**Keywords and visual cues**
Personal identity and sensitive information	Personal information	Identity information (e.g., name, resident registration number, passport number, driver's license number, face image, fingerprint, iris, and other biometric information); health information (e.g., height and weight, medical history, disability, health records, medication and injection information, prescription information, allergy information); account information (e.g., email address, username or ID, password, mobile phone number, SNS account, IP address); financial and asset information (e.g., bank account number, credit card number, salary statement, wage record, source of business or other income, salary or compensation history, credit rating or record, payment and overdue records, wage garnishment); online information (e.g., personal files exposed on a laptop or computer, internet browsing history, chat records); vehicle license plate.
Personal identity and sensitive information	Workplace / school	Employer or workplace name, company uniform, office interior, company logo, group photo, company dinner, employment certificate, school uniform, gym uniform, report card, exam paper, school record, learning or attendance record, student ID, diploma, award certificate, exam admission ticket, university logo.
Personal identity and sensitive information	Family events	Wedding invitation, wedding photograph, funeral hall, funeral portrait, funeral procession, obituary notice.
Location information	Public institutions and facilities	Public bathhouse, jjimjilbang, swimming pool, hospital, and similar public or institutional facilities.
Location information	Living space exposure	Home interior, kitchen, toilet, bathroom, bed, laundry, trash, and other domestic space cues.
Location information	Location cues	Home exterior, entrance, street address, signboard, school name, GPS, subway station name, road-name address, lot-number address, building sign, community service center, interior construction notice, welfare center, airline ticket information, railway ticket information, bus ticket information.
Body and appearance	Body exposure and sexual implication	Bare body, chest, buttocks, genitals, nudity, exposure, underwear, sexual pose, physical intimacy such as kissing, embarrassing posture, sexual activity, clothing that emphasizes body contours.
Body and appearance	Physical illness or injury	Facial injury, damaged facial skin, skin trouble such as acne, surgery marks or surgical sites, wounds, X-ray images.
Body and appearance	Appearance	Before-and-after cosmetic surgery, tattoo, piercing, body painting, sexually suggestive costume.
Socio-cultural sensitive information	Gender identity	LGBTQ+ symbols, rainbow flag, parade photograph, queer event, transgender identity, coming-out related content, homosexuality-related content.
Socio-cultural sensitive information	Religion / culture	Religious symbols such as cross, hijab, or monk; symbols of specific religions or denominations, including Christianity, Hinduism, and Islam; religious group activities; group proselytizing scenes; street preaching; group prayer; religious buildings such as churches and temples.
Socio-cultural sensitive information	Political or social position	Protest scene, political slogan, specific political party, picket sign, slogan text, leaflet, labor union membership, strike, political donation.
Socio-cultural sensitive information	Racial or ethnic origin	Skin color, racial appearance, ethnic traditional clothing, ethnic cultural symbols, multicultural expression, images expressing ethnic stereotypes.

Table 1: Full keyword-based privacy-relevant information taxonomy used for hashtag-based retrieval. The taxonomy includes six high-level privacy-relevant categories, subcategories, and keywords or visual cues. *Social relations* and non-sensitive public images were collected for dataset coverage and annotation but are not included in this retrieval-focused keyword table.

High-level category	Subcategory	Keywords and visual cues
Personal preferences	Private activity exposure	Drinking, alcohol such as vodka, soju, beer, wine, and makgeolli, clubbing, loss of consciousness, drunken state, cigarette, smoking, nightlife, nightclub, party clothing.
Personal preferences	Hobbies / leisure / style	Fashion style, game preference, travel preference, pet ownership, exercise preference, music genre, media consumption preference, hair dye, nail art, anime costume.
Risk and crime-related information	Violence / threat	Damage, scar, firearm, war, assault, weapon, self-harm, suicide, beating, harsh treatment, combat, shooting.
Risk and crime-related information	Crime / illegal / risky behavior	Theft, gambling, drugs, murder, abuse, offender attire, handcuffs, police dispatch, prison, arrest, body-shot, smuggler, robbery, child in danger, crime photograph.

Table 2: Full keyword-based privacy-relevant information taxonomy used for hashtag-based retrieval, continued.

A.2 Data Collection Details

Some taxonomy keywords could not be used for retrieval: certain suicide- and self-harm-related terms were restricted by Instagram's content-safety policies, while others (e.g., financial or institutional terms) yielded very few relevant images. During screening, we excluded images whose depicted content did not semantically match the target category, as well as duplicate, commercial or promotional images, and images unsuitable for analysis because of image quality or processing issues.

A.3 Detailed Scoring Criteria

This section summarizes the scoring anchors provided to annotators for each ordinal dimension. All criteria are defined with respect to the Korean social context, following the sensitive-information definitions of the Korean Personal Information Protection Act and the GDPR.

Identifiability measures the degree to which a specific individual can be directly or indirectly identified. Score 0 covers cases with no identifiable individuals, such as scenes with only objects or scenery, or people shown from behind or in silhouette. Score 1 covers partial or indirect identifiability, such as blurred or partially occluded faces, or partial information such as initials or a partly visible address. Score 2 covers clear identifiability, such as a clearly visible frontal face, identity documents, or personal information such as a home address, phone number, or vehicle license plate.

Location sensitivity measures how sensitive the depicted location is when disclosed, rather than merely whether it can be identified. Score 0 covers non-sensitive everyday spaces, such as parks, streets, cafes, restaurants, and tourist sites. Score 1 covers contextually sensitive places, such as workplace or school interiors and private indoor spaces. Score 2 covers locations under legal or strong cultural protection, such as hospitals and medical facilities, bathhouses and changing rooms, detention facilities, and military installations.

Activity sensitivity measures how sensitive the depicted or implied behavior is. Score 0 covers ordinary everyday activities, such as walking, eating, studying, or shopping. Score 1 covers behaviors that may cause contextual discomfort but are not strongly stigmatized, such as drinking or smoking. Score 2 covers behaviors associated with clear social stigma, legal risk, or serious privacy harm, such as sexual content, illegal activity, or self-harm. When only objects are present, the score reflects the activity they strongly evoke (e.g., an alcohol bottle implies drinking).

A.4 Annotation Schema

Table 3 summarizes the full annotation schema across the six components, including label types, scales, and the aggregation rule applied to each. The aggregation rules reflect the asymmetric risk of privacy underestimation: OR and MAX rules prioritize recall, while AND reflects the need for consensus on sharing decisions.

B Additional Model Results

B.1 Distribution of Sensitivity Scores

Figure 1 shows the distribution of sensitivity scores across the three annotation dimensions. Identifiability follows a bimodal pattern, while location and activity sensitivity are skewed toward low scores, indicating that high-sensitivity contextual cases are relatively infrequent in the curated dataset. Panel (b) shows the proportion of images with high location or activity sensitivity despite low or no identifiability.

B.2 Joint Distribution of Sensitivity Dimensions

Figure 2 shows the joint distributions of identifiability with location and activity sensitivity. Many images with high identifiability receive low location or activity sensitivity (e.g., 33.6% at identifiability = 2, location = 0), while a smaller set with low identifiability still receives high contextual sensitivity. This confirms that the three dimensions capture distinct information rather than collapsing into a single pattern.

Component	Type	Scale	Agg.	Interpretation
Information and observable attributes				
Information type	Multi-label	8 binary labels	OR	Type of privacy-relevant information present
Blur presence	Binary	0/1	OR	Intentional obfuscation applied by the uploader
Privacy sensitivity				
Identifiability	Ordinal	0–2	MAX	Degree to which an individual can be identified
Location sensitivity	Ordinal	0–2	MAX	Privacy harm of exposing the depicted location
Activity sensitivity	Ordinal	0–2	MAX	Stigma or legal risk of the depicted behavior
Sharing behavior				
Sharing allowance	Multi-label	6 tiers	AND	Perceived appropriate scope of sharing

Table 3: Annotation schema and aggregation rules. Each image is annotated along six components across three conceptual groups: information presence, privacy sensitivity (ordinal 0–2), and sharing behavior. Aggregation rules reflect the asymmetric risk of privacy underestimation: OR and MAX rules prioritize recall, while AND reflects the need for consensus on sharing decisions.

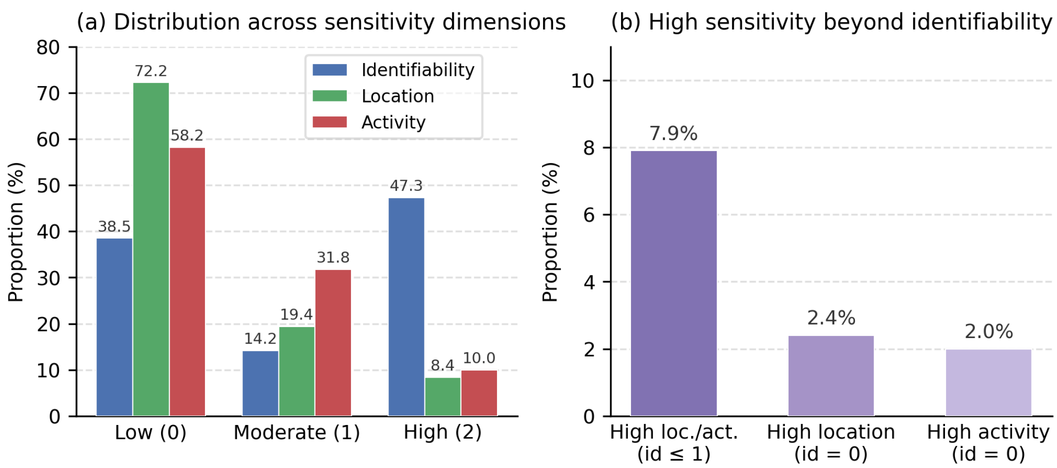

Figure 1: (a) Distribution of privacy sensitivity scores across the three annotation dimensions. (b) Proportion of images with high location or activity sensitivity despite low or no identifiability.

B.3 Sensitivity Profiles by Information Type

Figure 3 shows mean privacy sensitivity scores for each information type category. Identity- and social-relation-related information is more strongly associated with identifiability, whereas risk- and crime-related information and body or appearance information show higher activity sensitivity. Location sensitivity is more elevated for categories involving socially and institutionally delicate contexts.

B.4 Robustness of the Sharing–Sensitivity Association

The main paper reports that activity sensitivity is most strongly associated with no-sharing decisions ($\rho = 0.284$), using scores aggregated by the maximum. To verify that this finding is not an artifact of the aggregation rule or of combining the two annotators, we recompute the Spearman correlation between each sensitivity dimension and the no-sharing judgment, both for each annotator individually and under three aggregation rules. No-sharing is defined as the “not shared with anyone” response; at the aggregate level it requires both annotators to select this option, consistent with the main paper.

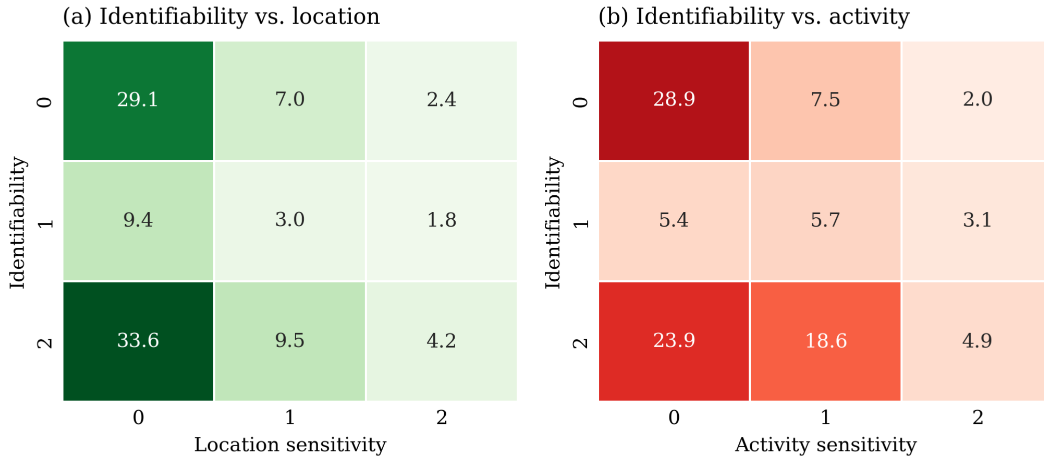

Figure 2: Joint distributions of identifiability with (a) location sensitivity and (b) activity sensitivity. Values indicate percentages over the full dataset.

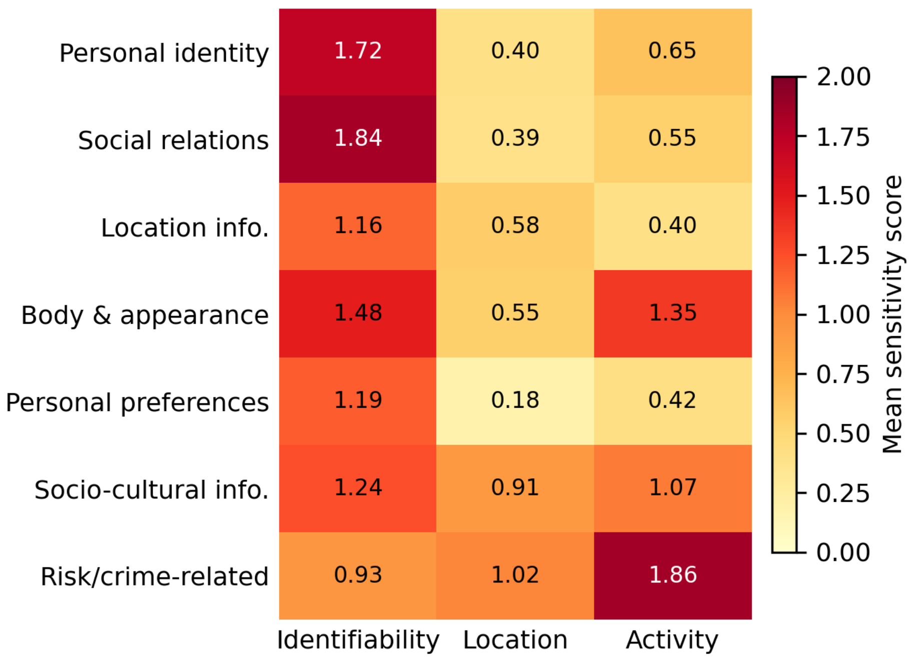

Figure 3: Mean privacy sensitivity scores across information types. Different categories of visual information are associated with distinct profiles across identifiability, location, and activity sensitivity.

Setting	**Identif.**	**Location**	**Activity**
Annotator 1 only	0.051	0.123	**0.274**
Annotator 2 only	0.127	0.151	**0.426**
Maximum (main)	0.090	0.132	**0.284**
Mean	0.074	0.127	**0.279**
Minimum	0.059	0.067	**0.258**

Table 5: Spearman correlations between each sensitivity dimension and no-sharing judgments, computed per annotator and under three aggregation rules ($p < .001$ throughout). Activity sensitivity shows the strongest association in every setting, confirming that the result is robust to both annotator choice and aggregation rule.

Group	Category / Dimension	κ	Agr.
Information type	Personal identity & sensitive information	0.832	0.919
	Socio-cultural sensitive information	0.800	0.958
	Risk/crime-related information	0.794	0.992
	Body and appearance	0.718	0.934
	Personal preferences	0.597	0.847
	Location information	0.511	0.850
	Social relations	0.505	0.910
	Group mean	*0.680*	*0.916*
Privacy sensitivity	Identifiability	0.865	0.814
	Activity sensitivity	0.723	0.820
	Location sensitivity	0.634	0.846
	Blur presence	0.844	0.988
	Group mean	*0.767*	*0.867*
Sharing allowance	Public	0.686	0.896
	Acquaintance	0.564	0.792
	Broadcast	0.494	0.912
	Normal relations	0.436	0.816
	No sharing	0.413	0.934
	Close relations	0.395	0.930
	Group mean	*0.498*	*0.880*

Table 4: Complete inter-annotator agreement across annotation components. Linearly weighted Cohen's κ is reported for ordinal sensitivity dimensions, and standard Cohen's κ for binary and multi-label components. Agr. denotes raw proportional agreement. The *other* category is excluded due to near-zero variance ($\kappa = 0.00$, raw agreement $= 0.906$).

B.5 Chance-Level Baselines

To contextualize the reported κ values, we compare against majority-class and random baselines under the same fold assignments and evaluation protocol as the main experiments. Random baselines are averaged over 100 draws. Both yield $\kappa \approx 0$ across all three dimensions, confirming that the learned models perform substantially above chance.

Baseline	Identif.	Location	Activity
Majority	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
Random (uniform)	0.001 ± 0.003	0.000 ± 0.002	0.001 ± 0.003
Random (label prior)	-0.001 ± 0.001	0.001 ± 0.003	-0.001 ± 0.002
Places365 + LogReg	0.664 ± 0.030	0.405 ± 0.045	0.419 ± 0.026
CLIP + MLP (best)	0.816 ± 0.017	0.647 ± 0.038	0.764 ± 0.010

Table 6: Chance-level baselines (majority and random) under the same protocol as the main experiments, with two learned models repeated for reference. All values are quadratic weighted κ.

B.6 Macro-F1 and Balanced Accuracy

The main paper reports quadratic weighted κ as the primary metric because the privacy sensitivity labels are ordinal. The tables below report additional results using macro F1 and balanced accuracy.

Representation	Classifier	Identifiability		Location		Activity	
		F1	Bal. Acc.	F1	Bal. Acc.	F1	Bal. Acc.
CLIP	LogReg	0.722 ± 0.011	0.731 ± 0.012	0.699 ± 0.024	0.734 ± 0.028	0.710 ± 0.010	0.726 ± 0.015
CLIP	Linear SVM	0.741 ± 0.015	0.747 ± 0.016	0.701 ± 0.017	0.721 ± 0.017	0.709 ± 0.017	0.719 ± 0.019
CLIP	MLP	0.755 ± 0.014	0.749 ± 0.010	0.758 ± 0.017	0.738 ± 0.022	0.771 ± 0.019	0.763 ± 0.023
DINOv2	LogReg	0.661 ± 0.012	0.667 ± 0.010	0.672 ± 0.020	0.690 ± 0.018	0.688 ± 0.021	0.697 ± 0.017
DINOv2	Linear SVM	0.658 ± 0.011	0.661 ± 0.009	0.642 ± 0.014	0.662 ± 0.011	0.679 ± 0.020	0.691 ± 0.018
DINOv2	MLP	0.726 ± 0.018	0.720 ± 0.021	0.746 ± 0.019	0.725 ± 0.020	0.747 ± 0.018	0.731 ± 0.019
Places365	LogReg	0.642 ± 0.009	0.645 ± 0.010	0.605 ± 0.023	0.621 ± 0.027	0.562 ± 0.013	0.567 ± 0.017
Places365	Linear SVM	0.633 ± 0.018	0.634 ± 0.018	0.579 ± 0.031	0.586 ± 0.035	0.533 ± 0.018	0.533 ± 0.021
Places365	MLP	0.661 ± 0.026	0.658 ± 0.024	0.627 ± 0.016	0.595 ± 0.017	0.616 ± 0.018	0.601 ± 0.015

Table 7: Additional frozen-representation classification results using macro F1 and balanced accuracy. Values are mean $\pm$ standard deviation over five-fold cross-validation.

Model	Identifiability		Location		Activity	
	F1	Bal. Acc.	F1	Bal. Acc.	F1	Bal. Acc.
CLIP FT	0.810 ± 0.011	0.819 ± 0.019	0.763 ± 0.020	0.747 ± 0.035	0.783 ± 0.013	0.784 ± 0.014
DenseNet-121 FT	0.718 ± 0.023	0.734 ± 0.028	0.596 ± 0.017	0.645 ± 0.029	0.600 ± 0.026	0.608 ± 0.030
EfficientNet-B0 FT	0.700 ± 0.016	0.715 ± 0.015	0.536 ± 0.011	0.632 ± 0.011	0.573 ± 0.011	0.603 ± 0.007

Table 8: Additional full fine-tuning results using macro F1 and balanced accuracy. Values are mean $\pm$ standard deviation over five-fold cross-validation. FT denotes full fine-tuning.

C Implementation Details

This section documents the hyperparameters, preprocessing, and cross-validation protocol used across all model evaluations, to support full reproducibility of the results reported in the main paper and Appendix B.

C.1 Full Fine-Tuning Configuration

Table 9 summarizes the training configuration used for CLIP ViT-B/16, DenseNet-121, and EfficientNet-B0. All three architectures were trained with a common protocol, differing only in the pretrained backbone, to support a fair comparison across representation families.

We use a smaller learning rate for the pretrained backbone than for the randomly initialized classification head, which reduces the risk of disrupting previously learned representations while still allowing task-specific adaptation. We did not perform architecture-specific hyperparameter optimization; instead, we used a standardized configuration across models to enable controlled comparison. Data augmentation was limited to horizontal flipping, which preserves privacy-relevant semantics such as identifiability, depicted location, and activity under a simple, architecture-agnostic transformation. Because location and activity sensitivity labels are right-skewed (72.1% and 58.1% of images are labeled 0, respectively), we apply class-balanced weighting to the training loss to prevent the majority class from dominating optimization.

C.2 Frozen-Representation Classifier Configuration

Table 10 reports the classifier hyperparameters used for the frozen-representation experiments (CLIP, DINOv2, and Places365 embeddings). All classifiers were evaluated using the same 5-fold `StratifiedGroupKFold`, with fold assignment constrained so that duplicate images detected from identical CLIP embeddings are placed in the same fold, preventing train–test leakage from duplicated visual content. We use exact-match detection rather than similarity thresholds; we verified that near-duplicate pairs (cosine similarity ≥ 0.98 between distinct embeddings) account for only 0.24% of the dataset, indicating negligible residual leakage.

The MLP classifier uses a shallow architecture to evaluate whether nonlinear decision boundaries can improve prediction from frozen representations, while avoiding excessive task-specific adaptation. Class balancing was also applied to logistic regression and linear SVM classifiers to address the same imbalanced label distributions; the MLP classifier is omitted from this setting because scikit-learn's `MLPClassifier` does not support per-class loss weighting.

C.3 Image Preprocessing

Each pretrained representation used its own standard preprocessing recipe, consistent with its original training procedure, rather than a single shared pipeline:

Hyperparameter	Value (shared across all three models)
Backbone learning rate	1×10^{-5}
Classification head learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Optimizer	AdamW
Learning rate schedule	Cosine annealing (no warmup)
Batch size	32
Maximum epochs	10
Early stopping patience	3 epochs (based on validation quadratic weighted κ)
Data augmentation	Random horizontal flip ($p = 0.5$)
Class weighting	Weighted cross-entropy loss (per-fold class weights via scikit-learn's `compute_class_weight(class_weight='balanced')`)
Pretrained weights	CLIP: `openai/clip-vit-base-patch16`; DenseNet-121 and EfficientNet-B0: ImageNet-1K
Random seed	42
Cross-validation protocol	5-fold `StratifiedGroupKFold`, with fold assignment constrained so that duplicate images detected from identical CLIP embeddings are placed in the same fold, preventing train–test leakage from duplicated visual content. Within each fold, 10% of the training-validation split is held out for early stopping and model selection; the test fold is used only for final evaluation.

Table 9: Training configuration for full fine-tuning experiments. All three architectures share the same optimization protocol, augmentation strategy, and class-weighting procedure; only the pretrained backbone differs.

Classifier	Configuration
Logistic Regression	`max_iter=3000`, `class_weight=balanced`, standardized input features
Linear SVM	`max_iter=2000`, `class_weight=balanced`, `dual=False`, standardized input features
MLP	Single hidden layer with 256 units, ReLU activation, Adam solver, ℓ_2 penalty $\alpha = 1 \times 10^{-4}$, batch size 64, initial learning rate 1×10^{-3}, maximum 200 iterations, early stopping enabled (validation fraction 0.1, 10 iterations without improvement)

Table 10: Classifier configurations for the frozen-representation experiments. All classifiers use scikit-learn default settings except where noted.

- **CLIP**: Resize and center-crop to 224×224 with bicubic interpolation, normalized using CLIP-specific channel statistics.
- **DINOv2**: Resize to 256×256 (bicubic interpolation) followed by a 224×224 center crop, normalized using standard ImageNet channel statistics.
- **Places365**: Resize to 256×256 (bilinear interpolation) followed by a 224×224 center crop, normalized using standard ImageNet channel statistics.

Each representation's native preprocessing pipeline was retained rather than standardized, since forcing a shared pipeline (e.g., applying CLIP-specific normalization to DINOv2) would introduce an input distribution mismatch relative to each model's pretraining setup, degrading representation quality and undermining the frozen-representation comparison. All embeddings were L2-normalized prior to classification.

C.4 Computing Infrastructure

All experiments were run on a single NVIDIA RTX 4090 GPU. The software environment used Python 3.11, PyTorch 2.10.0 with CUDA 12.8, torchvision 0.25.0, transformers 5.6.2, and scikit-learn 1.8.0.